Thiago Ferreira dos Santos

Physicist & Astrophysicist | 4th-year PhD candidate | Numerical Modelling, Simulations, and Data Analysis

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Education

Yale University (US) – PhD Candidate in Astrophysics	2023–Present
Universidade de São Paulo (Brazil) – MSc in Astronomy	2021–2023
Universidade Federal de Santa Catarina (Brazil) – BSc in Physics	2016–2021

Technical Skills

Programming & Computing: Python (10+ years experience); Yale high performance computing (HPC) clusters, batch job scheduling;
Simulation & Modelling: Modules for Experiments in Stellar Astrophysics (MESA; 1D stellar evolution), GYRE (stellar oscillations);
Galactic Dynamics & Population Synthesis: Gala, Galpy (orbit-integration software), COSMIC (stellar binary population synthesis);
Data Analysis: Gaia, TESS, VVV-X (astronomical survey datasets), high-resolution spectroscopy, radial velocities, chemical abundances, Gaussian Process regression (statistical modelling under uncertainty);
Observational Instrumentation (major ground-based telescopes/spectrographs): Keck/HIRES, Gemini/GHOST & MAROON-X, Subaru/HDS, VLT/ESPRESSO & HARPS, Magellan/MIKE, PFS, WINERED.

Technical Experience

Yale University – PhD Researcher in Astrophysics — 2023–Present

- Building and analysing MESA/GYRE stellar model grids across low-mass, near-zero-metallicity regimes, systematically varying overshoot efficiencies, mass-loss prescriptions, semi-convective and convective boundary mixing to characterise their effects on interior stellar structure, evolution, nucleosynthetic yields, and asteroseismic (mixed g-mode/p-mode) oscillation spectra of metal-free/Population III stars—the first asteroseismic study for this stellar population—resulting in 2 first-author refereed papers.
- Building Python pipelines using Gala/Galpy to cross-match *Gaia* DR3 astrometry, TESS photometry, and high-resolution spectroscopy into homogeneous multi-survey samples, computing orbital integrals of motion (energy, eccentricity, angular momentum) for kinematic classification of the Milky Way halo substructure, feeding directly into an in-progress paper testing a stellar stream association (a debris trail from a long-ago merged dwarf galaxy) for a newly identified halo eclipsing binary, and now extended to more than double the Galactic thick-disk planet-host sample for a follow-up study of thick disc rocky-exoplanet bulk densities.
- Serving as Principal Investigator on 12 nights and Co-Investigator on 20+ additional nights of competitively awarded telescope time across 8 major observatories, owning proposal development, technical planning, target selection, high-resolution echelle spectroscopic data acquisition, and joint transit+radial-velocity Keplerian orbital modelling, including a 5-night, 3-observatory PI campaign (Keck/HIRES, Subaru/HDS, Gemini/GHOST) uncovering early supernova nucleosynthesis signatures in carbon-enhanced metal-poor (CEMP) halo giant stars, and a 6-night PI survey constraining the occurrence rate of Solar-System-like giant-planet architectures around Sun-like stars (ESO/HARPS).
- Authored 7 first-author and co-authored 8 peer-reviewed publications spanning stellar evolution, exoplanets, brown dwarfs, white dwarfs, and Galactic stellar populations, including spin-orbit alignment and tidal-eccentricity characterisation of transiting brown dwarfs (2 first-author papers), a benchmark white dwarf–ultracool dwarf wide binary, and hierarchical Bayesian eccentricity modelling of thick-disk planetary systems; referee for The Astrophysical Journal.

Universidade de São Paulo – MSc Researcher in Astronomy — 2021–2023

- Detected 16 exoplanetary systems around solar-twin stars from ESO/HARPS radial-velocity time series (Solar Twin Planet Search programme), separating genuine planetary signals from stellar-activity-induced radial velocity variations via activity-indicator and timing analysis, and applying Keplerian and quasi-periodic Gaussian Process models to characterise orbits, uncovering planetary engulfment signatures and outer-companion architectures, including several Neptune-like and Jupiter-analogue planets.

Selected Achievements

GSAS Yale Graduate Fellowship	2023–Present
Boris Garfinkel Prize Fellowship in Astronomy	2023
Invited Seminar, HITS/Heidelberg, Germany	Jul 2025
MESA Summer School, Leuven, Belgium	2025
CAPES M.Sc. Research Fellowship, IAG/USP, Brazil	2021–2023

Teaching & Communication

Languages: Portuguese (native), English (fluent), Spanish (working proficiency);
Teaching Assistant, Yale University, 2024–2026 (ASTR 110, 220, 310, and 556) and IAG/USP, 2022 (AGA0502).